

Figure 1

R code to Read the British Airway Reviews Dataset

```
# Load required package
library(readr)

test<- read_csv("/Users/mrinalikapalaniappan/Desktop/SUSS/Y2S1/ANL312/ECA/British_reviews_skytrax.csv")
```

Figure 2

Output after the dataset was read

```
> # Load required package
> library(readr)
> test<- read_csv("/Users/mrinalikapalaniappan/Desktop/SUSS/Y2S1/ANL312/ECA/British_reviews_skytrax.csv")
New names:
• `` -> `...1`
Rows: 3400 Columns: 2
— Column specification —
Delimiter: ","
chr (1): reviews
dbl (1): ...1

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Figure 3

R code used for Sampling the dataset

```
#####Sampling#####  
  
# Load the dataset  
BA <- read_csv("/Users/mrinalikapalaniappan/Desktop/SUSS/Y2S1/ANL312/ECA/British_reviews_skytrax.csv")  
  
#Set a random seed for reproducibility  
set.seed(123)  
  
#Randomly sample 600 reviews (ensuring > 500 rows)  
sample_BA <- BA[sample(1:nrow(BA), 600, replace = FALSE), ]  
  
#Saving the sampled dataset as a csv file  
write_csv(sample_BA, "/Users/mrinalikapalaniappan/Desktop/SUSS/Y2S1/ANL312/ECA/British_reviews_sample.csv")  
  
#Verify output  
dim(sample_BA) |
```

Figure 4

R code for Quality Check

```
#Data Quality Check  
  
sum(is.na(sample_BA)) #checking for missing entries  
sum(duplicated(sample_BA)) #checking for duplicate entries
```

Figure 5

R code for Text Pre-Processing

```
#####Text Pre-Processing#####
# Load libraries
library(topicmodels)
library(slam) #install slam if package not installed
library(SnowballC)
library(tm)

#Read the sampled dataset
txt <- read.csv("/Users/mrinalikapalaniappan/Desktop/SUSS/Y2S1/ANL312/ECA/British_reviews_sample.csv",
               encoding = "latin1", stringsAsFactors = FALSE)

#Prepare dataset for VCorpus
colnames(txt) <- c("id", "text") # first column = id, second = review text
txt$doc_id <- 1:nrow(txt) # unique document ID
txt <- txt[, c("doc_id", "text")] # reorder for DataframeSource

#Convert to corpus
comments_corpus <- VCorpus(DataframeSource(txt))
comments <- comments_corpus

#Cleaning steps
comments <- tm_map(comments, content_transformer(tolower)) # lowercase
comments <- tm_map(comments, removePunctuation) # remove punctuation
comments <- tm_map(comments, removeNumbers) # remove numbers

# keep only alphabets & spaces (DO NOT remove spaces!)
toSpace <- content_transformer(function(x, pattern) gsub(pattern, " ", x))
comments <- tm_map(comments, toSpace, "[^A-Za-z\\s']")

#Custom stopwords for British Airways reviews
mystopwords <- scan(file = "/Users/mrinalikapalaniappan/Desktop/SUSS/Y2S1/ANL312/SU4/mystopwords.data",
                   what = "character", sep = "\n")

# Combine standard + custom stopwords (moderate removal)
custom_stopwords <- c(mystopwords[1:671], stopwords("english"),
                     "nil", "na", "trip", "verified", "review", "reviews", "flight", "flights",
                     "airline", "airlines", "aircraft", "ba", "british", "airways", "airway", "plane",
                     "flew", "flying", "fly", "flown", "travel", "traveller", "travelling", "travelled",
                     "customers", "customer", "passenger", "passengers", "people")

comments <- tm_map(comments, removeWords, custom_stopwords)

#Final text normalization
comments <- tm_map(comments, stemDocument) # stemming
comments <- tm_map(comments, stripWhitespace) # remove extra spaces
```

Figure 6

R code for Document Term Matrix and Term frequency-inverse document frequency

```
##### Document Term Matrix #####
dtm<-DocumentTermMatrix(comments)
inspect(dtm) # inspect the DTM
dtm_tfidf<-weightTfidf(dtm, normalize=TRUE)
inspect(dtm_tfidf)

# Convert to matrix
m_tfidf <- as.matrix(dtm_tfidf)

#Compute mean tf-idf per term
term_means <- colMeans(m_tfidf)

#Compute threshold as median
threshold <- median(term_means)

#Select terms above threshold: keep the terms whose mean tf-idf is ≥ the threshold and extracts the names (terms) of the elements that passed the threshold
selected_terms <- names(term_means[term_means >= threshold])

#Subset original DTM (not the tf-idf one)
dtm_reduced <- dtm[, selected_terms] # only selected terms in DTM are retained
dtm_reduced<-dtm_reduced[row_sums(dtm_reduced)>0,] # keep the documents with at least one term

# inspect the reduced document-term-matrix and check the dimension
inspect(dtm_reduced)
dim(dtm_reduced)
```

Figure 7

R code for Document Term Matrix and Term frequency-inverse document frequency

```
##### LDA #####
# Define topic range
k_values <- 4:10

# Create storage
perplexities_vem <- numeric(length(k_values))
loglik_vem <- numeric(length(k_values))
perplexities_gibbs <- numeric(length(k_values))
loglik_gibbs <- numeric(length(k_values))

# Loop to calculate perplexity and log-likelihood
for (i in seq_along(k_values)) {
  k <- k_values[i]

  # ----- VEM -----
  lda_vem <- LDA(dtm_reduced, k = k, method = "VEM", control = list(seed = 123))
  perplexities_vem[i] <- perplexity(lda_vem, dtm_reduced)
  loglik_vem[i] <- logLik(lda_vem)

  # ----- Gibbs -----
  lda_gibbs <- LDA(dtm_reduced, k = k, method = "Gibbs",
    control = list(seed = 123, burnin = 1000, thin = 100, iter = 1000))
  perplexities_gibbs[i] <- perplexity(lda_gibbs, dtm_reduced)
  loglik_gibbs[i] <- logLik(lda_gibbs)
}
```

Figure 8

R code for Perplexity and Log-likelihood for VEM and Gibbs

```
##### LDA #####
# Define topic range
k_values <- 4:10

|
perplexities_vem <- numeric(length(k_values))
loglik_vem <- numeric(length(k_values))
perplexities_gibbs <- numeric(length(k_values))
loglik_gibbs <- numeric(length(k_values))

# Loop to calculate perplexity and log-likelihood
for (i in seq_along(k_values)) {
  k <- k_values[i]

  # VEM
  lda_vem <- LDA(dtm_reduced, k = k, method = "VEM", control = list(seed = 123))
  perplexities_vem[i] <- perplexity(lda_vem, dtm_reduced)
  loglik_vem[i] <- logLik(lda_vem)

  #Gibbs
  lda_gibbs <- LDA(dtm_reduced, k = k, method = "Gibbs",
                  control = list(seed = 123, burnin = 1000, thin = 100, iter = 1000))
  perplexities_gibbs[i] <- perplexity(lda_gibbs, dtm_reduced)
  loglik_gibbs[i] <- logLik(lda_gibbs)
}

##### Perplexity and Log-Likelihood for VEM #####

quartz()
# Perplexity (VEM)
plot(k_values, perplexities_vem, type = "b", pch = 19, col = "red",
     xlab = "Number of Topics (k)", ylab = "Perplexity",
     main = "Perplexity vs Number of Topics (VEM)")
grid()

quartz()
# Log-Likelihood (VEM)
plot(k_values, loglik_vem, type = "b", pch = 19, col = "red",
     xlab = "Number of Topics (k)", ylab = "Log-Likelihood",
     main = "Log-Likelihood vs Number of Topics (VEM)")
grid()

##### Perplexity and Log-Likelihood for Gibbs #####

# Perplexity (Gibbs)
quartz()
plot(k_values, perplexities_gibbs, type = "b", pch = 19, col = "blue",
     xlab = "Number of Topics (k)", ylab = "Perplexity",
     main = "Perplexity vs Number of Topics (Gibbs)")
grid()

# Log-Likelihood (Gibbs)
quartz()
plot(k_values, loglik_gibbs, type = "b", pch = 19, col = "blue",
     xlab = "Number of Topics (k)", ylab = "Log-Likelihood",
     main = "Log-Likelihood vs Number of Topics (Gibbs)")
grid()
```

Figure 9

R code for LDA models built for k values of 4 to 8

```
##### Manual Analysis of K value #####

k<-4 #generate 4 topics
seed<-123
model_lda<-list(VEM=LDA(dtm_reduced,k=k,method="VEM",control=list(seed=seed)),
               Gibbs=LDA(dtm_reduced,k=k,method="Gibbs",control=list(seed=seed,burnin=1000,thin=100,iter=1000)))

#results
terms_vem<-terms(model_lda$VEM,10) # top 10 terms in each topic of VEM model
terms_gibbs<-terms(model_lda$Gibbs,10) # top 10 terms in each topic of Gibbs model
terms_vem
terms_gibbs

k<-5 #generate 5 topics
seed<-123
model_lda<-list(VEM=LDA(dtm_reduced,k=k,method="VEM",control=list(seed=seed)),
               Gibbs=LDA(dtm_reduced,k=k,method="Gibbs",control=list(seed=seed,burnin=1000,thin=100,iter=1000)))

#results
terms_vem<-terms(model_lda$VEM,10) # top 10 terms in each topic of VEM model
terms_gibbs<-terms(model_lda$Gibbs,10) # top 10 terms in each topic of Gibbs model
terms_vem
terms_gibbs

k<-6 #generate 6 topics
seed<-123
model_lda<-list(VEM=LDA(dtm_reduced,k=k,method="VEM",control=list(seed=seed)),
               Gibbs=LDA(dtm_reduced,k=k,method="Gibbs",control=list(seed=seed,burnin=1000,thin=100,iter=1000)))

#results
terms_vem<-terms(model_lda$VEM,10) # top 10 terms in each topic of VEM model
terms_gibbs<-terms(model_lda$Gibbs,10) # top 10 terms in each topic of Gibbs model
terms_vem
terms_gibbs

k<-7 #generate 7 topics
seed<-123
model_lda<-list(VEM=LDA(dtm_reduced,k=k,method="VEM",control=list(seed=seed)),
               Gibbs=LDA(dtm_reduced,k=k,method="Gibbs",control=list(seed=seed,burnin=1000,thin=100,iter=1000)))

#results
terms_vem<-terms(model_lda$VEM,10) # top 10 terms in each topic of VEM model
terms_gibbs<-terms(model_lda$Gibbs,10) # top 10 terms in each topic of Gibbs model
terms_vem
terms_gibbs

k<-8 #generate 8 topics
seed<-123
model_lda<-list(VEM=LDA(dtm_reduced,k=k,method="VEM",control=list(seed=seed)),
               Gibbs=LDA(dtm_reduced,k=k,method="Gibbs",control=list(seed=seed,burnin=1000,thin=100,iter=1000)))

#results
terms_vem<-terms(model_lda$VEM,10) # top 10 terms in each topic of VEM model
terms_gibbs<-terms(model_lda$Gibbs,10) # top 10 terms in each topic of Gibbs model
terms_vem
terms_gibbs
```

Figure 10

R code for LDA models built for k values of 9 to 10

```
k<-9 #generate 9 topics
seed<-123
model_lda<-list(VEM=LDA(dtm_reduced,k=k,method="VEM",control=list(seed=seed)),
               Gibbs=LDA(dtm_reduced,k=k,method="Gibbs",control=list(seed=seed,burnin=1000,thin=100,iter=1000)))

#results
terms_vem<-terms(model_lda$VEM,10) # top 10 terms in each topic of VEM model
terms_gibbs<-terms(model_lda$Gibbs,10) # top 10 terms in each topic of Gibbs model
terms_vem
terms_gibbs

k<-10 #generate 10 topics
seed<-123
model_lda<-list(VEM=LDA(dtm_reduced,k=k,method="VEM",control=list(seed=seed)),
               Gibbs=LDA(dtm_reduced,k=k,method="Gibbs",control=list(seed=seed,burnin=1000,thin=100,iter=1000)))

#results
terms_vem<-terms(model_lda$VEM,10) # top 10 terms in each topic of VEM model
terms_gibbs<-terms(model_lda$Gibbs,10) # top 10 terms in each topic of Gibbs model
terms_vem
terms_gibbs
```

Figure 11

R code for GammaDF table and the Top 5 Reviews for Each Topic

```
##### Evalaution #####
k<-6 #generate 6 topics
seed<-123
model_lda<-list(VEM=LDA(dtm_reduced,k=k,method="VEM",control=list(seed=seed)),
               Gibbs=LDA(dtm_reduced,k=k,method="Gibbs",control=list(seed=seed,burnin=1000,thin=100,iter=1000)))

#results
terms_vem<-terms(model_lda$VEM,10) # top 10 terms in each topic of VEM model
terms_gibbs<-terms(model_lda$Gibbs,10) # top 10 terms in each topic of Gibbs model
terms_vem
terms_gibbs

gammaDF_gibbs<-as.data.frame(model_lda$Gibbs@gamma,row.names=model_lda$Gibbs@documents)
names(gammaDF_gibbs)<-c(1:k)
View(gammaDF_gibbs)

write.csv(gammaDF_gibbs, "/Users/mrinalikapalaniappan/Desktop/SUSS/Y2S1/ANL312/ECA/gammaDF_gibbs.csv", row.names = TRUE)

#Sort and Retrieve the Top 5 reviews per Topic
for (topic in 1:k)
{print(paste("Top 5 reviews for Topic", topic))
  top_docs <- order(gammaDF_gibbs[, topic], decreasing = TRUE)[1:5]
  print(txt$text[top_docs])
  print("-----")}}
```